

DATABASE SCHEMA

SQL-HR-Analytics-Project

PostgreSQL Database Documentation

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Version 1.1 (Updated — missing columns added)

Overview

This document describes the complete PostgreSQL database schema for the SQL-HR-Analytics-Project. It lists every table, column, data type, constraint, and description used across the HR database, including fields added after the initial design pass (marked "added" below).

Database Tables

1. Departments Table

Stores the organizational departments that employees belong to.

Column	Data Type	Constraint	Description
department_id	SERIAL	PRIMARY KEY	Unique department identifier
department_name	VARCHAR(100)	NOT NULL	Department name
location	VARCHAR(100)	NULL	Department location (added)

2. Jobs Table

Stores job titles along with their minimum and maximum salary bands.

Column	Data Type	Constraint	Description
job_id	SERIAL	PRIMARY KEY	Unique job identifier
job_title	VARCHAR(100)	NOT NULL	Job title
min_salary	NUMERIC(10,2)	NULL	Minimum salary for the role (added)
max_salary	NUMERIC(10,2)	NULL	Maximum salary for the role (added)

3. Employees Table

Stores employee personal details, employment status, compensation, and reporting hierarchy.

Column	Data Type	Constraint	Description
employee_id	SERIAL	PRIMARY KEY	Unique employee identifier
fname	VARCHAR(50)	NOT NULL	First name
lname	VARCHAR(50)	NOT NULL	Last name
email	VARCHAR(100)	UNIQUE	Employee email
phone_number	VARCHAR(20)	NULL	Employee phone number (added)
gender	VARCHAR(10)	NULL	Employee gender (added)
dob	DATE	NULL	Date of birth (added)

Column	Data Type	Constraint	Description
department_id	INT	FOREIGN KEY	References Departments
job_id	INT	FOREIGN KEY	References Jobs
manager_id	INT	SELF FOREIGN KEY	References Employees (reporting manager)
basic_salary	NUMERIC(10,2)	NOT NULL	Employee salary
hire_date	DATE	NOT NULL	Joining date
status	VARCHAR(20)	DEFAULT 'Active'	Employment status (added)

Relationships

Departments (1) — (Many) **Employees**

Jobs (1) — (Many) **Employees**

Employees (Manager) (1) — (Many) **Employees**

A department can have many employees, but each employee belongs to exactly one department. The same one-to-many pattern applies between Jobs and Employees. The Employees table also has a self-referencing relationship: `manager_id` points back to `employee_id`, so a manager (one row in Employees) can supervise many other employees (many other rows in the same table).

Changes From Initial Schema

The following columns were missing from the original schema and have been added in this version:

Departments — *location*

Jobs — *min_salary, max_salary*

Employees — *phone_number, gender, dob, status*